Supplement to "Patch-Aware KV Cache Compression for Byte-Level Language Models"

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Extra charts and the analytic compute table that did not fit in the 10-page report. Every chart is parsed from `results/part3/*.md` by `report/make_figures.py`; the table is generated from `results/part3/flops.md`.

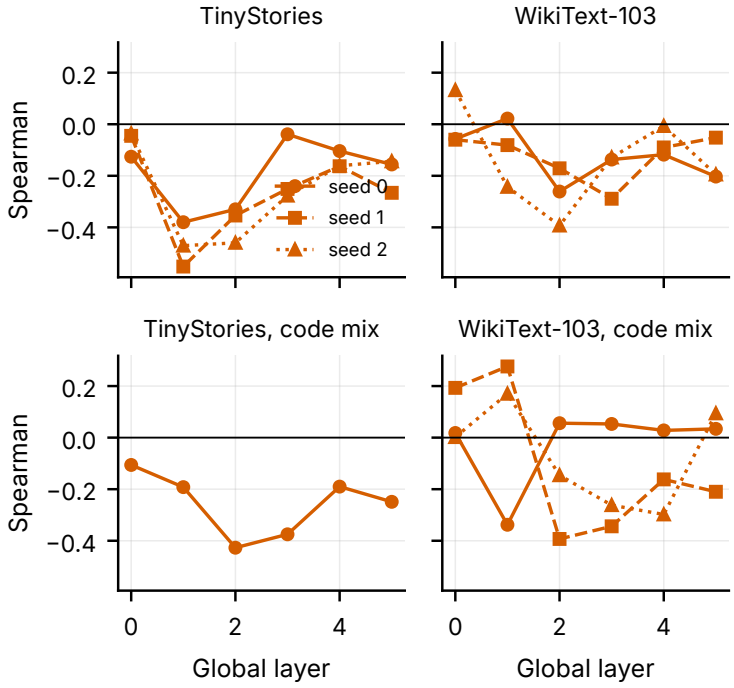


Figure S1. Spearman correlation between the entropy of an old patch (at least 8 patches back) and the attention it receives, per global layer and seed. It is negative in the middle and upper layers of most models: the model reads predictable patches more than surprising ones. Correlational evidence only; the WikiText code-mix seed 0 model is the outlier.

Table S1. Analytic MFLOPs per generated byte and KV size (MB, fp16) at two contexts. The BPE Transformer was trained for 256 tokens (about 1,024 bytes), so its 4,096-byte rows are hypothetical. Formulas and assumptions in `results/part3/flops.md`.

	1,024 bytes		4,096 bytes	
Model	MFLOPs	MB	MFLOPs	MB
BLT 13.3M, full global cache	11.06	2.1	12.46	8.4
BLT, 64-patch cache	10.73	0.6	10.73	0.6
BLT, full cache, plus entropy model	19.58	2.1	20.98	8.4
BLT, 64-patch cache, plus entropy model	19.25	0.6	19.25	0.6
BPE transformer 13.8M, full cache	7.43	2.4	9.20	9.4
BPE transformer, 64-token cache	6.99	0.6	6.99	0.6
Byte-level transformer, full cache	30.87	9.4	59.18	37.7
Byte-level transformer, 64-entry cache	22.02	0.6	22.02	0.6
Byte-level transformer, 288-entry cache (same text span as 64 patches)	24.08	2.7	24.08	2.7

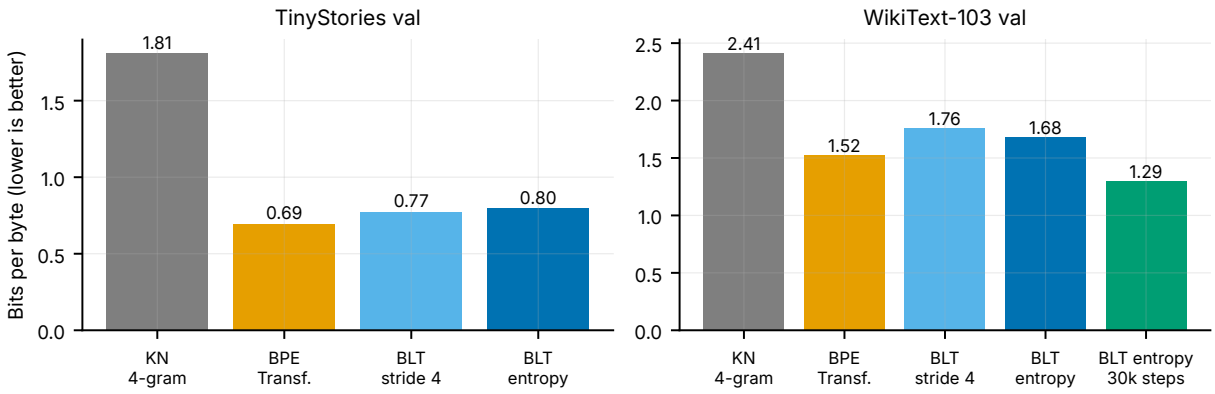


Figure S2. Bits per byte of the classical baseline (Kneser-Ney byte 4-gram), the modern baseline (BPE Transformer) and the BLT-style models. At equal training (3,000 steps) the BPE Transformer is ahead; with ten times more training the WikiText code-mix BLT is lower (a comparison that gives the BLT more training and a code mix).

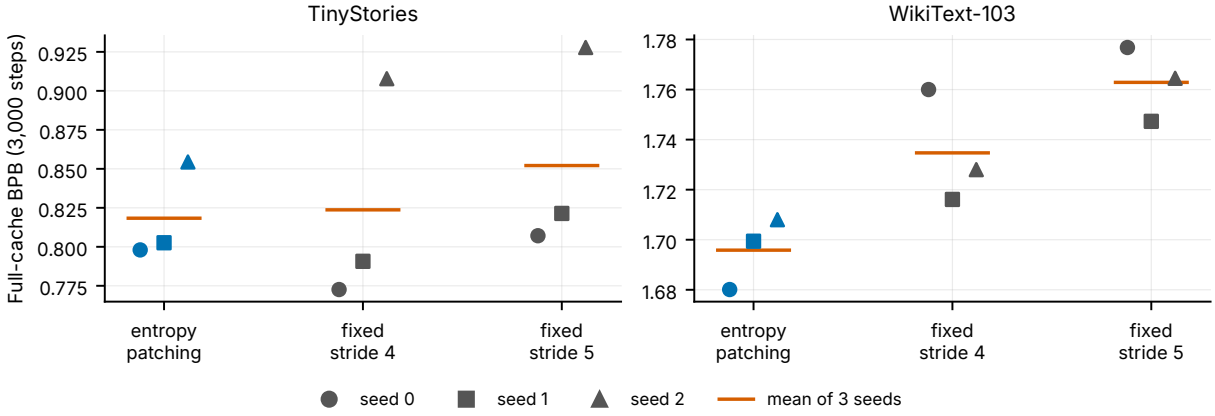


Figure S3. Why we ran three seeds. Full-cache BPB for every seed of each patching scheme: the seed spread (0.03 to 0.14) is as large as the gaps between schemes, so single-seed patching comparisons are not reliable. Entropy patching is lowest on WikiText in all three seed pairs.

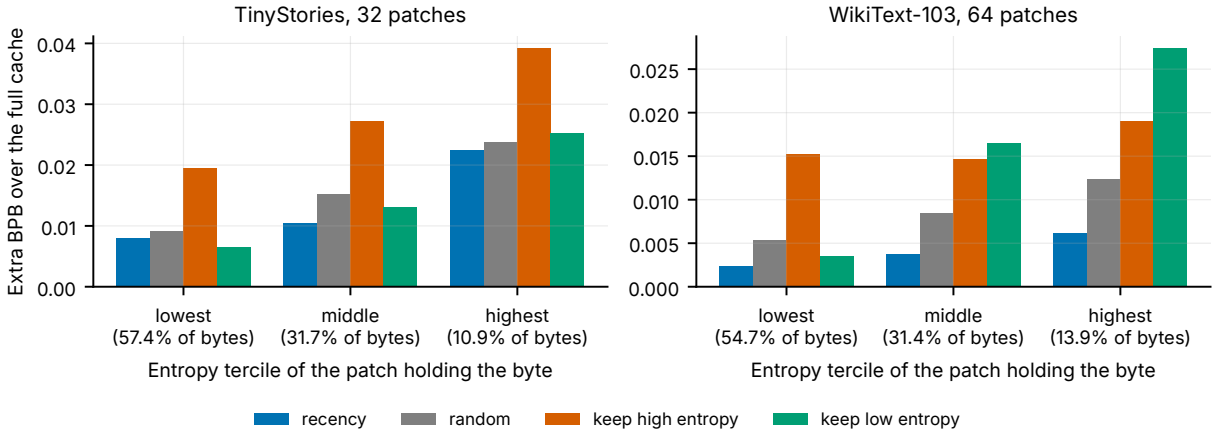


Figure S4. Extra loss over the full cache, by the entropy tercile of the patch holding the byte (first patch and 8 newest protected). Keeping high-entropy patches does not protect the high-entropy bytes and also costs the easy bytes.